

Assignment 1

Due Date 27 March 2026 (5:20 pm)

Covers material:

- i. Biased and mean square error
- ii. Confidence intervals for one population mean
- iii. Testing of hypotheses for one population mean
- iv. Confidence intervals for two population means
- v. Testing of hypotheses for two population means

Involves:

Doing all work by handwritten and scanned in as a pdf and please upload to BB

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Q1: Let $Y_1, Y_2, \dots, Y_n$ be a random sample with $E(Y_i) = \mu$ and $V(Y_i) = \sigma^2$.

Show that

- i.* $\bar{Y}$ is an unbiased estimator of population mean μ .
- ii.* S^2 is an unbiased estimator of population variance σ^2 .

use,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

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Q2: Researchers are concerned about the impact of students working while they are enrolled in classes. A survey of 200 students provides a sample mean of 7.10 hours a week worked, with the standard deviation is about 3 hours.

- i. What is a 95% confidence interval for population mean?
- ii. At 5% level of significance test the hypotheses that students worked more than 7 hours a week.

(Find the value of the test statistic, find the rejection region for the test, Conduct the test and state your conclusions.)

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Q3: Independent random samples of $n_1 = 29$ and $n_2 = 18$ observations were selected from two normal populations with unequal variances and found $\bar{Y}_1 = 37.6, \bar{Y}_2 = 35.9, S_1 = 5.2$ and $S_2 = 6.7$.

Suppose you wish to detect a difference between the population means.

- i. State the null and alternative hypotheses for the test.
- ii. Find the rejection region for the test use $\alpha = 0.01$.
- iii. Find the value of the test statistic.
- iv. Calculate the approximate p-value for the test.
- v. Conduct the test and state your conclusions.